

Locomotion-Grounded Humanoid Soccer: Task-Gated Reinforcement Learning of a Multi-Directional Kicking Library

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Abstract—Recent humanoid soccer systems make motion tracking the substrate and derive locomotion from it, typically by steering a motion-reference anchor toward the ball. This yields strong shooting results, but locomotion is trained only on the narrow, deterministic command distribution ball approach induces, never evaluated as a capability in its own right. We invert the stack: a general, command-conditioned locomotion policy is trained first as the substrate, and N motion-guided kicking skills are added on top as task-gated layers, so the reachable gait space is set by the locomotion curriculum rather than any reference clip. Because every skill starts from and returns to this same commandable state, locomotion also becomes a composition hub ($O(N)$ transitions rather than $O(N^2)$), and post-strike stabilisation is handed back to the trained controller rather than scripted per clip. We instantiate this on a 29-DoF Unitree G1 with seven retargeted kicking skills spanning 259.5° of nominal aim direction, including lateral, rearward and weak-foot strikes a single forward-facing reference cannot express, and report shooting accuracy alongside command-tracking, terrain and push-recovery results with the full skill library attached, an axis prior humanoid soccer systems do not report. The library is validated on hardware across forward, lateral, rearward and commanded approaches.

I. INTRODUCTION

Soccer stresses two humanoid whole-body control capabilities at once, on one controller and the same hardware. The first is sustained, general locomotion: repositioning, retreating, sidestepping, turning in place. The second is precise, contact-rich striking, whose useful signal is a sub-10 ms impulse with delayed feedback about where the ball went. Reinforcement learning has advanced both halves [1]–[7]. The strongest reported shooting result on a full-size humanoid attains 0.73 m mean target error from 3 m on hardware, at peak ball speeds of 13.10 m/s [2].

The recipe behind these results is consistent across systems. A human kicking clip is retargeted to the robot [8], [9]; a tracking reward supplies the whole-body coordination pure task rewards struggle to discover [10]–[12]; ball and target rewards are layered on top. Locomotion enters last and indirectly, by steering the tracking anchor toward the ball while the tracking reward stays active throughout approach [2]. Motion tracking is the *substrate*.

This ordering, whose reward machinery we largely adopt, has two consequences the literature has not measured. First, locomotion is exercised only during ball approach, so the training command distribution is whatever that heuristic emits. The anchor-steering law of [2] issues no lateral, backward or in-place-rotation command, and no command-tracking, rough-terrain, push-recovery or omnidirectional

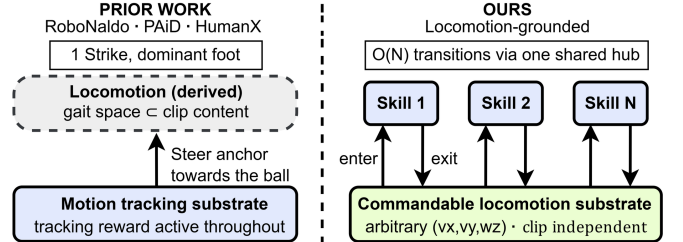


Fig. 1: **Substrate inversion.** Prior work derives locomotion from a ball-steered anchor; ours is the commandable base.

result is reported there or in [3], [4]. That much is curriculum, not architecture, since Stage-3 locomotion in [2] is already reward-driven. What is architectural is whether the tracking objective is switched off or merely discounted while locomotion trains (Sec. III-A). Either way, general locomotion in a humanoid soccer policy is neither trained broadly nor measured. That is the gap this paper closes.

Second, training on a single reference kick bounds the repertoire to one contact type. RoboNaldo [2] reports a right-footed bias, degradation at extreme lateral and high targets, and names multi-skill priors an open problem. A match calls for a *library*: an inside-foot pass, a lateral flick, a strike behind the current heading.

Approach. We invert the stack (Fig. 1). A general, command-conditioned locomotion policy is trained first, with a randomised (v_x, v_y, w_z) envelope, mixed terrain, push/mass/friction/actuation randomisation, and no ball or reference clip present. N motion-guided kicking skills are then added as task-gated layers, hard-zeroed in locomotion mode rather than discounted (Fig. 2), so the reachable gait space is set by the locomotion curriculum alone. Every skill’s post-strike recovery hands back to this same controller, in place of the synthetic, momentum-blind stabilisation tail prior work relies on: without that tail, RoboNaldo’s alive rate falls 98.8%→24.4%.

We instantiate this on a 29-DoF Unitree G1 at 50 Hz. Training runs in IsaacLab [13] with an off-policy maximum-entropy learner [14], [15], and per-skill specialists are merged into one deployable network by DAgger-style distillation [16], [17] (Fig. 2). We evaluate in MuJoCo [18] sim-to-sim on the exported policy and on hardware.

Contributions. (i) **A locomotion-grounded architecture** that hard-zeroes kick objectives outside kick mode and attaches the skill library at a locomotion cost of at most 0.04 in command-tracking MAE (Table XI). (ii) **A multi-skill strike**

library spanning 259.5° of nominal aim, including lateral, rearward and weak-foot strikes, reported per-skill and as a union with gaps included. (iii) **A learned kick→locomotion handoff** in place of a scripted stabilisation tail, with an interruptibility analysis of survival when a skill is aborted mid-swing. (iv) **A joint evaluation of locomotion generality and shooting quality**: command tracking, terrain traversal and push recovery with the library attached, a combination not reported elsewhere.

II. RELATED WORK

A. Humanoid Soccer and Robot Sports

On bipeds, Haarnoja et al. [1] demonstrated end-to-end one-versus-one soccer on a miniature platform, with walking, turning, kicking and getting up, and transitions learned rather than scripted. Its policy is conditioned on egocentric game state rather than a velocity command, and it learns a single scoring strike rather than a library.

On full-size humanoids, a growing line targets shooting directly: [19] optimises foot trajectories without learning; [7] addresses goalkeeping under position-conditioned constraints; [6] and [4] study vision-driven and noisy-perception striking; [3] proposes a progressive perception-action framework; [5] learns generalisable interaction from human video. Concurrently, SkillX [20] composes trapping, dribbling and shooting in one policy, selected by a skill command rather than a velocity command. It transitions skill-to-skill with no locomotion state between, and shoots at a goal rather than a commanded aim. RoboNaldo [2] is the closest comparison, reporting the strongest shooting numbers via a three-stage curriculum: clip imitation, ball and target rewards, then a locomotion command interface. We adopt much of its reward design. Multi-skill soccer policies exist; what is unreported is locomotion measured independent of ball approach, and directional strike coverage rather than a set of distinct behaviours. Table I scores these axes (✓ supported, ◦ partial, ✗ absent; multi-skill counts ball-interaction skills, not strike directions). No prior system earns a full mark on free command, omnidirectional locomotion or rearward targets.

B. Motion Tracking, Locomotion and Object Interaction as Primitives

Reference-guided RL supplies whole-body coordination that task rewards alone struggle to discover [10], [11]. Recent humanoid systems track retargeted motion directly on hardware [8], [9], [12], the pipeline we use for every clip. Tracking supplies stability but constrains timing and contact selection, a tension staged curricula aim to relax [2]. Our task-mode gating relaxes it fully, switching the tracking objective off rather than discounting it outside the skill. Separately, massively parallel simulation has made velocity-commanded locomotion with terrain and disturbance curricula well established [13], recently at minutes of wall-clock cost via off-policy maximum-entropy learning [14], [15]. We place this at the *base* of the stack, not as a ball-approach by-product. The contribution is learner-agnostic: task gating, the hub and the handoff are properties of the formulation,

not the optimiser. Contact-rich loco-manipulation usually assumes dwell-time contact, with reward accumulated over many steps [21]. Shooting is the opposite: a brief impulse whose reward arrives only after the ball has travelled. That motivates the aiming rewards used here and in [2], and explicit handling of post-contact stability.

C. Skill Libraries, Composition and Transitions

The options framework formalises temporally extended actions via initiation sets, policies and termination conditions [22]. A shared hub state makes every skill’s termination lie in the next’s initiation by construction, reducing transitions from $O(N^2)$ to $O(N)$. That is the role our locomotion mode plays. We merge specialists into one network via behaviour distillation [16], [17]: independently trained teachers supervise one student under its own state distribution. Haarnoja et al. merge differently, training under self-play with a KL regulariser that keeps the policy near the skills; ours regresses directly onto a routed teacher’s actions, with no reward term in merging. A single deployed network is not itself distinguishing: both [1] and [2] deploy one non-hierarchical network. What differs is which capability it is grounded in, and how many strike types it carries.

III. PROBLEM FORMULATION

We formulate the problem as a single task-conditioned Markov decision process $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ solved by one policy π_θ . Each environment additionally carries a task-mode variable

$$m_t \in \mathcal{M} = \{\text{LOCO}, \text{KICK}_1, \dots, \text{KICK}_N\}, \quad (1)$$

selecting, at every timestep, which reward and termination specification the transition is evaluated against. This is the mechanism behind the *free command* and *omnidirectional* columns, where Table I gives no prior system a full mark. $\mathcal{M}$ contains LOCO as an ordinary mode, not a residual of ball approach.

A. Task-mode gating

Rather than arbitrate between separate controllers, m_t gates reward, termination and observation inside one network. With $\mathcal{T}$ indexing reward terms and w_τ the weight of term τ ,

$$r_t = \sum_{\tau \in \mathcal{T}} g_\tau(m_t) w_\tau \hat{r}_\tau(s_t, a_t), \quad g_\tau : \mathcal{M} \rightarrow \{0, 1\}, \quad (2)$$

an indicator equal to 1 only in the mode τ was authored for. The gate is a hard zero, not a reduced weight: in locomotion mode every kick term contributes nothing to the gradient, and conversely. This is what carries the substrate claim, and Table X(c) measures it. Discounting the tracking terms rather than zeroing them costs command tracking, and the cost grows with the discount: backward MAE rises 0.12 at a $0.1\times$ floor, the weight [2] itself applies in locomotion mode, and 0.33 at $0.5\times$, the latter worse at all eight matched checkpoints (sign test $p=0.008$). All four command axes degrade monotonically. Terminations are gated identically.

The observation is gated by masking, not omission: inactive-mode terms are zeroed but stay in the vector, so

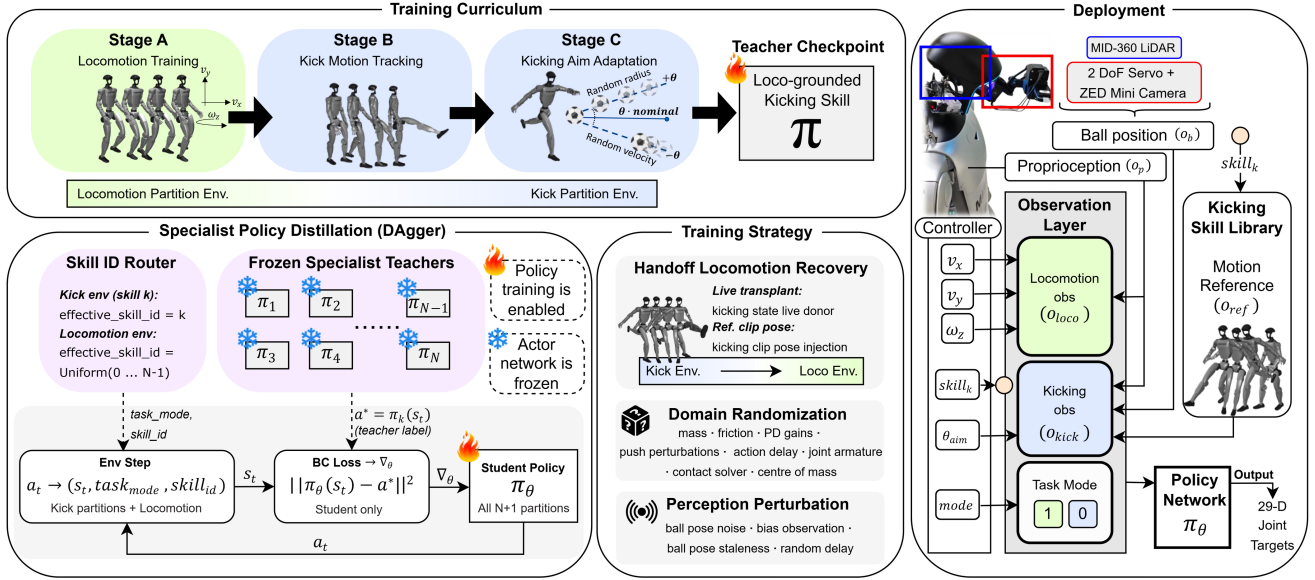


Fig. 2: **Training pipeline and deployed architecture.** **Training (left).** Stages A–C are Stages 1–3 of Sec. IV: locomotion first and alone, then motion tracking, then motion-guided kicking, with specialist distillation (Stage 4) below. **Deployment (right).** The same distilled, task-mode-gated network, with the perception hardware of Sec. V supplying ball position.

TABLE I: Capability comparison against prior humanoid soccer systems. Lateral and rearward targets require the strike bearing to be an explicit command input spanning that range; systems that reach such bearings only emergently score $\circ$.

Method	Platform	Locomotion substrate		Skill repertoire				Handoff	
		Free command	Omni directional	Multi skill	Motion guided	Lateral target	Rearward target	Loco $\rightarrow$ Kick	Kick $\rightarrow$ Loco
Haaroja et al. [1]	Miniature	$\times$	$\circ$	$\times$	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$
STOFT [19]	Bipedal	$\times$	$\times$	$\checkmark$	$\times$	$\checkmark$	$\times$	$\checkmark$	$\checkmark$
Reactive [6]	Humanoid	$\times$	$\times$	$\circ$	$\checkmark$	$\circ$	$\circ$	$\checkmark$	$\circ$
Striker [4]	Humanoid	$\circ$	$\times$	$\times$	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$
Goalkeeper [7]	Humanoid	$\times$	$\times$	$\checkmark$	$\checkmark$	$\times$	$\times$	$\times$	$\times$
HumanX [5]	Humanoid	$\times$	$\times$	$\checkmark$	$\checkmark$	$\times$	$\times$	$\times$	$\times$
PAiD [3]	Humanoid	$\times$	$\times$	$\checkmark$	$\checkmark$	$\times$	$\times$	$\checkmark$	$\times$
RoboNaldo [2]	Humanoid	$\circ$	$\times$	$\times$	$\checkmark$	$\circ$	$\times$	$\checkmark$	$\circ$
SkillX [20]	Humanoid	$\times$	$\times$	$\checkmark$	$\checkmark$	$\circ$	$\times$	$\times$	$\times$
Ours	Humanoid	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark^{\S}$	$\checkmark$

^{$\S$} Survival stays above 95% while entry accuracy falls with the speed carried into the strike (Sec. VI-C).

the input is 261-dimensional in every mode (Table II). A loco/kick one-hot is never masked, so the policy is told its mode rather than inferring it from which blocks are zero.

B. Skill conditioning

Within kick mode a skill is specified two ways. Its retargeted reference trajectory enters the observation directly (Table II), making the policy reference-conditioned rather than index-conditioned: there is no learned skill embedding. The strike direction is *commanded*, not targeted directly as a world-frame point. Given the ball’s realised planar position p_{ball} (itself randomised at reset) and a skill’s nominal bearing β , the target is synthesised as

$$p_{\text{tgt}} = p_{\text{ball}} + D [\cos(\beta + \theta), \sin(\beta + \theta)]^T, \quad (3)$$

with $D=5$ m for every skill and θ sampled once per attempt. Only θ reaches the policy, normalised as

$$\theta_{\text{aim}} = \theta / \theta_{\text{ref}} \in [-1, 1], \quad (4)$$

with $\theta_{\text{ref}}=45^\circ$ a fixed constant, so the command is bounded by construction and needs no robot localisation at deployment. It occupies a two-wide observation slot whose second component is reserved and held at zero (Table II).

Eq. 3 is what separates this from targeting a fixed world-frame point, as in [2]’s 8×2 m goal plane 5 m ahead. Wherever reset randomisation placed p_{ball} , $p_{\text{tgt}} - p_{\text{ball}}$ is exactly D in direction $\beta + \theta$, so θ alone carries the aim error. A world-frame point target cannot give this: its scale and reachable range are set by each skill’s own geometry. Here $\theta_{\text{aim}} \in [-1, 1]$ is one bounded command axis for all N .

C. Handoff

A skill ends when its authored reference content is exhausted, m_t returns to LOCO, and the trained locomotion policy resumes control. We call this a handoff rather than a stabilisation phase because no synthetic recovery reference

is tracked. The robot is handed back to a disturbance-robust controller in whatever single-support, momentum-carrying state the strike left it. The reverse transition, Loco→Kick, is commanded rather than learned, with the precondition Table I footnotes: entry accuracy falls with the velocity carried into the strike. The simulated sweep also places the ball rather than walking to it (Sec. VI-C).

Table II details the observation space of Fig. 2’s deployment view; every term is robot-measurable except base linear velocity, which is critic-only.

TABLE II: Actor observation space, 261 dimensions.

<i>Locomotion group (o_{loco})</i>		<i>Kicking group (o_{kick})</i>	
Term	Dim	Term	Dim
Base angular velocity	3	Motion reference ($q, \dot{q}$)	58
Projected gravity	3	Reference orientation	6
Command (v_x, v_y, ω_z)	3	Base angular velocity	3
Joint position	29	Joint position	29
Joint velocity	29	Joint velocity	29
Previous action	29	Previous action	29
Gait phase, sin	2	Ball position o_b	3
Gait phase, cos	2	Aim command θ_{aim}	2
Task-mode one-hot, active in both modes: 2			

IV. METHOD

Training proceeds in the four stages of Fig. 2. The ordering is the claim: locomotion is trained first and alone, and everything else is layered onto it.

A. Stage 1: locomotion substrate

A command-conditioned locomotion policy is trained with no ball and no reference clip present, on commands sampled from a randomised (v_x, v_y, ω_z) envelope covering forward, backward, lateral and in-place rotation, over mixed terrain with disturbance randomisation. This stage fixes the reachable gait space, and with no clip present that space is a property of the command curriculum alone.

B. Stage 2: motion tracking

Kick-partition environments add a tracking objective against a human kick clip, recovered from video [8] and retargeted [9]. Ball and target rewards stay at zero weight here, so the policy learns the strike’s coordination before it is asked to aim. Locomotion environments continue the unmodified Stage-1 task: this is an addition, not a replacement.

C. Stage 3: motion-guided ball kicking

The shooting objective ramps linearly from zero over 5000 iterations, held at 0 throughout Stage 2. The aim angle θ of Eq. 3 is sampled per attempt within $\pm 15^\circ$ of the skill’s nominal bearing, with ball placement randomised by ± 10 cm; Table III lists the main activated reward terms (each group additionally scaled: tracking $2\times$, shooting $0.5\times$, others $1\times$), with the alive reward attenuated $10\times$ before contact so standing still is not viable.

Balance-potential shaping. The nine $\exp(-e^2/\sigma^2)$ tracking terms of Table III saturate toward 0 exactly where error is diverging toward a fall, and the flat per-step alive bonus

TABLE III: Kick-mode reward terms and weights. Twenty-two standard safety and regularisation terms (weights -100 to 50) are omitted for space.

Term	w	Term	w
<i>Motion tracking</i>			
Global root pos.	1.0	Global root orient.	0.5
Relative body pos.	1.0	Relative body orient.	1.0
Global body lin. vel.	1.0	Global body ang. vel.	1.0
Global feet lin. vel.	1.0	Strike joint null-sp.	1.0
Strike leg null-sp.	1.0		
<i>Shooting</i>			
Ball-to-target error	20.0	Ball velocity	10.0
Predicted ball-to-tgt.	5.0	Contact orientation	4.0
Ball proximity	2.0	CoM to ball dist.	2.0
Torso to ball dist.	2.0	Foot strike pitch	1.0
Ball over line	1.0	Weak foot contact	-1.0
Self contact (feet)	-0.4		
<i>Post-strike recovery posture</i>			
Stand height	-40	Stance asymmetry	-20
Stand orientation	-20	Feet width	-20
Knee width	-20	Yaw drift	-2
<i>Alive and balance shaping</i>			
Kick alive	4.0	Balance shaping (Eq. 5)	1.0

has zero action-gradient by construction, leaving the pre-fall regime without a usable signal. We add potential-based shaping over a capture-point balance margin,

$$F(s, s') = \gamma \Phi(s') - \Phi(s), \quad \Phi(s) \in [0, 1], \quad (5)$$

with $\Phi = \Phi_{\text{bal}}\Phi_{\text{ht}}$ a balance margin (capture-point distance to the support polygon) times a fall-height margin. Ng et al. [23] show this form leaves the optimal policy unchanged for any Φ and weight, provided γ matches the agent’s discount and Φ is zeroed only at a true termination; unlike every other weight in Table III, which trades stability against kick quality, this one can be set aggressively, and no analogous shaping is reported in [2]’s otherwise-similar rewards.

TABLE IV: Skill library geometry.

Skill	Foot	Nominal bearing	Reachable band ($\theta_{\text{max}} = \pm 15^\circ$)
skill_1	right	-1.32°	$[-16.3, +13.7]$
skill_2	right	$+137.80^\circ$	$[+122.8, +152.8]$
skill_3	left	-121.74°	$[-136.7, -106.7]$
skill_4	right	-56.94°	$[-71.9, -41.9]$
skill_5	right	$+48.39^\circ$	$[+33.4, +63.4]$
skill_6	right	0.00°	$[-15.0, +15.0]$
skill_7	right	$+43.89^\circ$	$[+28.9, +58.9]$
Nominal span 259.5° ; reachable union 155.8° over five islands			

Handoff boundary training. Some locomotion environments are seeded at reset from a kick-disturbed pose, taken from the reference clip or, with higher probability, a live running kick environment’s own state. The latter is faithful where the former is not, since tracking error is never zero and contact forces have no motion-capture representation. The seeding is enabled during distillation (Stage 4); the ablation retrains it at specialist level to isolate it (Table X(b)).

D. Stage 4: specialist distillation

Each skill is trained as above from the shared Stage-1 substrate, one locomotion-grounded specialist per clip. The specialists are then merged into one deployable actor π_θ by

TABLE V: Shooting quality in context. PPO, AMP and PAiD rows are the free-kick simulation baselines as reported in [2].

Method	Bearing range	Dist. (m), tol.	suc.@0.5 m (%)	suc.@1.0 m (%)	Shot err. (m)	$v_{\text{ball}}^{\text{max}}$ (m/s)	Hit (%)	Alive (%)
PPO [24]	forward	n/r	0.0±0.0	0.0±0.0	4.721±.007	0.780±.003	0.0±0.0	0.0±0.0
AMP [11]	forward	n/r	0.9±0.1	3.3±0.1	3.733±.004	1.771±.051	38.6±2.8	41.6±2.3
PAiD [3]	forward	n/r	8.2±0.7	19.5±0.5	1.850±.026	4.986±.017	67.7±0.3	67.3±0.5
RoboNaldo [2], stage 2, free-kick	forward, ±38.7°	5.0, n/r	28.8±1.3	65.5±0.6	0.899±.012	14.792±.011	82.1±0.3	100.0±0.0
RoboNaldo [2], stage 3, free-kick	forward, ±38.7°	5.0, n/r	36.3±0.1	66.5±0.2	1.055±.012	14.178±.034	81.1±0.1	99.7±0.1
RoboNaldo [2], stage 3, moving-ball	forward, ±38.7°	5.0, n/r	32.4±0.6	63.3±0.6	1.131±.004	13.875±.004	79.2±0.1	98.8±0.1
Ours , loco-grounded specialist, skill_1	≈forward, ±15°	5.0, ±5.7°	44.3±16.8	77.5±12.1	0.69±.17	3.67±.15	100.0±0.0	100.0±0.0
<i>Multi-directional library: a broader capability envelope (Table I)</i>								
Ours , unified $N=7$, library	259.5° span, 155.8° union	5.0, ±5.7°	15.4±12.7	32.2±21.9	1.68±.76	2.86±.77	96.7±5.9	99.4±1.0

DAGger-style distillation [16], [17]:

$$\min_{\theta} \mathbb{E}_{s \sim d_{\pi_{\theta}}} [\mathcal{L}(\pi_{\theta}(s), \pi^{(k(s))}(s))], \quad (6)$$

where $k(s)$ is the skill-index router selecting which frozen teacher labels state s : a kick-partitioned environment keeps its own specialist through its locomotion ticks, and a purely locomotion-partitioned one draws a skill index uniformly over the N specialists. $d_{\pi_{\theta}}$ is the state distribution induced by the *student's own* rollout, not the teachers' trajectories, which is what separates Eq. 6 from behaviour cloning: supervision covers the states the student drifts into.

Two details are load-bearing: each teacher must run under its own frozen observation normaliser (the wrong one yields a confident wrong action and raises nothing), and the rollout action form must match the regression target, since sampling one and regressing the other is self-amplifying.

E. Learning algorithm

All stages use FastSAC, an off-policy maximum-entropy actor-critic [14], [15], adopted because simulation throughput, not environment count, is the bottleneck in contact-rich training, so replay extracts more updates per simulated step. Running two tasks in one network motivates a mode-dependent discount and entropy target, $\gamma(m_t)$ and $\alpha(m_t)$, keyed to m_t as in Eq. 2, so the horizon suited to a multi-second kick is not forced onto locomotion's shorter one.

F. Domain randomisation

Randomisation broadens once at the Stage-1 to Stage-2 boundary, then holds constant. Beyond mass, centre-of-mass, friction, PD-gain and action-delay randomisation, two axes target the strike specifically: joint armature with internal friction, and the contact solver's own response parameters. Standard humanoid randomisation leaves how contact is *resolved* fixed throughout, which a policy can overfit for free. The ball observation is perturbed on its own axes (additive noise, a fixed per-episode bias, delayed and held samples, a stale-reading probability), so it never sees a perfect estimate.

V. EXPERIMENTAL SETUP

A. Platform, simulation and skill library

All experiments use a 29-DoF Unitree G1, policy at 50 Hz emitting joint position targets, trained in IsaacLab [13] with thousands of parallel environments. Every simulated performance number comes from the exported deployment policy running in MuJoCo [18], not from training telemetry,

so the cross-engine transfer gap sits inside the measurement rather than behind it. On hardware, ball position is supplied onboard by a head-mounted MID-360 LiDAR and a 2-DoF servo-mounted ZED Mini camera (Fig. 2).

Seven kicking skills, each retargeted from a separate human video clip, are summarised in Table IV: each nominal bearing follows from the skill's configured ball and target placement (0° straight ahead, positive left), and the reachable band is that bearing $\pm \theta_{\text{max}}$, with $\theta_{\text{max}} = 15^\circ$.

B. Metrics and protocol

Shooting success is scored at 0.5 m and 1.0 m radius (nearest post-contact ball-to-target distance; a whiff counts as failure), alongside shot error, peak ball speed and contact rate. With every target at $D=5$ m (Eq. 3), a 0.5 m success allows $\pm 5.7^\circ$ of aim for every skill. Stability uses a strict fall criterion, not the tracking-based alive flag. Locomotion generality is command-tracking error, terrain traversal and push recovery, all with the library attached.

Unified-policy numbers are medians over a checkpoint sweep (8 checkpoints, 16 for $N=7$, 250–300 attempts each), not single readings: single-checkpoint evaluation is unreliable (one skill ranged 3.6–45.2% across a 16-checkpoint sweep). Teacher *success* is not swept, since each teacher is the fixed distillation artifact. We report medians, not means, since per-checkpoint distributions are zero-heavy and right-skewed. All results are single training runs except Table XI's baseline, which is measured over three seeds.

VI. RESULTS

Teacher entries are the single checkpoint each student was distilled from; every other number follows Sec. V.

A. Shooting quality

Table V compares against prior systems on the metrics they report (bold = best; n/r = not reported; Hit = ball-foot contact, RoboNaldo's Contact). The $\pm$ differs by row: seed spread as published for prior systems, scan spread for our specialist, across-skill spread for our library. Point accuracy is rarely reported: of the systems in Table I, only [2] scores shots by distance to a target point, the metric we adopt. Within our own table, only the upper, forward-cone block is like-for-like; the lower, multi-directional row is a different capability envelope, not a worse entry in the same contest. At comparable forward distance the skill_1 specialist reaches 44.3% success at 0.5 m, with standard error 1.7 over 100 scans. That is the highest figure we are aware of on this

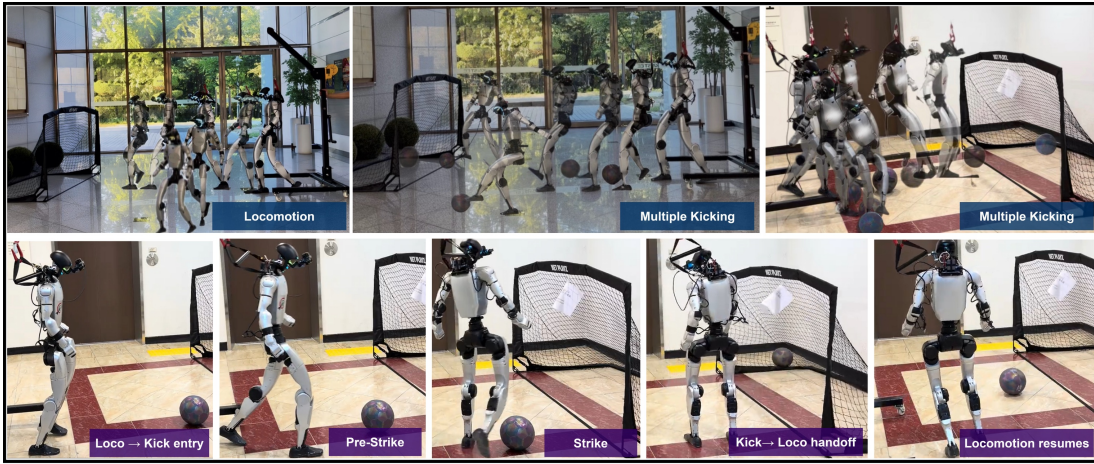


Fig. 3: **Hardware experiments on the Unitree G1.** **Top:** commanded locomotion and two library strikes, each a multi-exposure composite of one trial. **Bottom:** a kick-to-locomotion handoff, from the mode switch through contact to locomotion resuming with no scripted stabilisation tail (Sec. VI-C).

metric, but it is not a like-for-like win: the aim cone is $\pm 15^\circ$ against $\pm 38.7^\circ$, the ball leaves at roughly a quarter of the pace, and it is one training run. The distilled seven-skill policy does not retain it: skill_1 falls to 10.2%, and the library mean is 15.4%.

Peak ball speed is the one column prior work dominates, and the gap is an operating point rather than a ceiling. In the five teachers with training logs, shooting earns 10–15% of the positive kick-mode reward and motion tracking 76–85%. Raising the shooting weight several-fold in an earlier configuration collapsed sim-to-sim survival outright, since strike power and destabilising contact force are physically coupled, not merely reward-coupled. The library is tuned for directional coverage at a survivable strike, not for pace.

Table IX(a) tests the library on hardware (Fig. 3). Skills are grouped by bearing from forward: skill_1 and skill_6 forward, skill_4 and skill_5 lateral, skill_2 and skill_3 rearward; skill_7 overlaps skill_5 and was not run separately. A commanded-approach condition exercises the locomotion-to-kick handoff under operator velocity commands. The test room allows only 1.42–2.08 m to the target, against $D=5$ m in simulation. Success therefore scales the 0.5 m radius to the nearest target, $r = 0.5 d_{\min}/D = 0.14$ m with $d_{\min} = 1.42$ m, so no trial is allowed more than the simulated $\pm 5.7^\circ$.

B. Kicking skill coverage

Table VI reports the final unified policy’s per-skill performance, medians over 16 checkpoints; Fig. 4 shows the library geometrically. Panel (a) plots every ball contact from 100 trials per skill (564 of 700 attempts; ball jittered ± 20 cm, twice the training range). The seven skills resolve into five islands, with skills 1/6 and 5/7 overlapping. Post-contact ball speed varies twofold (2.04–4.13 m/s, Table VI) with no forward-versus-lateral pattern. Trajectories spill outside their $\pm 15^\circ$ aim bands, which is the aim bias Sec. VII quantifies rather than commanded spread. Panel (b) sweeps 1500 placements over a region spanning all seven spawns, assigning each to its nearest skill (70.4% contact overall). Of

these, 78.5% lie outside every skill’s ± 10 cm training box. Contact survives displacement far better than aim does: only 11.9% of placements pass within 1.0 m of the target.

Topple rates stay low across all seven bearings (mean 0.6%, worst 2.6%), but aim does not: success spans 0.8% to 36.6%, and **interference from merging skills is the dominant cause**. Table VII compares each skill’s success against the teacher it was distilled from, across four cumulative library sizes. Mean success declines monotonically from 26.0% at $N=2$ to 15.4% at $N=7$, a deficit already present at the smallest library and widening with every skill added. The merge is therefore the bottleneck: every specialist was trained from the same Stage-1 base. skill_2 is the clearest case: distilled from one of the weakest teachers (19.6%) into the most extreme rotation ($+137.8^\circ$), it contacts the ball 92.0% of the time yet collapses to a 0.8% median. skill_1 and skill_6 contact on every attempt yet land only 10.2% and 7.6%. All three are aiming failures rather than striking ones: a bearing correction alone lifts them to 20–29% (Table XII).

TABLE VI: Per-skill performance of the unified policy.

Skill	1	2	3	4	5	6	7	mean
Suc.@0.5 (%)	10.2	0.8	20.6	25.8	36.6	7.6	6.4	15.4
Suc.@1.0 (%)	22.0	3.2	40.4	51.0	67.4	23.8	17.8	32.2
Shot err. (m)	1.90	2.92	0.98	1.10	0.81	1.89	2.15	1.68
$v_{\text{ball}}^{\text{max}}$ (m/s)	3.63	2.22	2.84	2.82	2.04	2.36	4.13	2.86
Hit (%)	100.0	92.0	85.2	100.0	100.0	100.0	100.0	96.7
Alive (%)	99.6	100.0	98.6	100.0	100.0	100.0	97.4	99.4

TABLE VII: Negative transfer: specialist vs. unified.

Skill	Teacher	$N=2$	$N=4$	$N=6$	$N=7$
skill_1	44.3	37.8 (−6.5)	22.5 (−21.8)	29.0 (−15.3)	10.2 (−34.1)
skill_2	19.6	14.2 (−5.4)	8.0 (−11.6)	5.3 (−14.3)	0.8 (−18.8)
skill_3	31.8	—	31.8 (+0.0)	19.2 (−12.6)	20.6 (−11.2)
skill_4	30.0	—	20.7 (−9.3)	8.8 (−21.2)	25.8 (−4.2)
skill_5	46.4	—	—	30.2 (−16.2)	36.6 (−9.8)
skill_6	41.8	—	—	15.3 (−26.5)	7.6 (−34.2)
skill_7	18.4	—	—	—	6.4 (−12.0)
mean	33.2	26.0 (−5.9)	20.8 (−10.7)	18.0 (−17.7)	15.4 (−17.8)

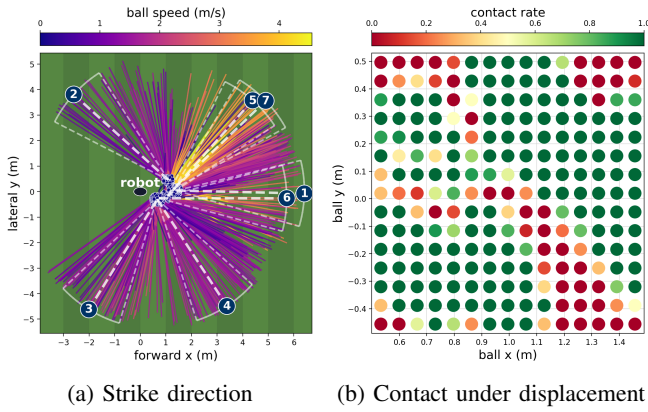


Fig. 4: What the library covers, and how far the ball may sit from where a skill expects it.

C. Handoff

Table VIII evaluates the final unified policy (medians over 16 checkpoints) on both directions of the mode boundary. Kick-to-locomotion, scored on survival alone, survives both the policy’s own auto-return and an abort at an arbitrary tick, including inside the strike window. Locomotion-to-kick sweeps entry speed from zero to the full locomotion range and is scored on survival, contact and aim, since contact alone saturates near 100% on most skills. **Entering the kick while moving costs both stability and accuracy, and the cost grows smoothly with entry speed.** Scaling the envelope moves contact 95.3→84.1, success at 0.5 m 17.0→11.9, at 1.0 m 32.7→25.7, and topple 0.7→4.2, monotonically on all four. Survival is not the fragile part: 95.8% of full-speed entries survive, against 99.3% from rest. Accuracy is: success at 0.5 m falls by nearly a third. The approach itself is unmeasured, since the ball is placed at its spawn rather than walked to.

Table X isolates the two mechanisms behind this robustness on skill_7 specialist checkpoints, not the deployed library. (a) ablates post-strike stabilisation, scored on settled-stance error rather than survival, since all three arms already survive at or above 99%. Our arm holds the lowest error and is the only one that settles at all. Contact safety without the posture term is worse than neither, so the two are not separable. (b) ablates the kick-state pose injection that trains locomotion recovery from live, kick-disturbed states rather than idealised reference-clip poses. The gap between an arm’s ref.-clip and live-donor columns widens monotonically the further training sits from the live distribution: 6.6 points for the transplant arm, 22.6 for reference-clip-only, 39.7 for none. The transplant arm’s live-donor 81.8% is not comparable to Table VIII’s pooled 98.8%. Measured the same strike-window-only way on the deployed checkpoints it is 98.1%, so the gap is checkpoint maturity, not disagreement. Table IX(b) confirms the main result on hardware: kick-to-locomotion survival under a mid-swing abort, triggered manually and scored phase-stratified rather than by a scripted random tick.

TABLE VIII: Handoff robustness, both directions.

Direction	Trigger condition	Alive	Hit	Suc.@0.5	Suc.@1.0
Kick→Loco	end of authored clip	99.1	n/a	n/a	n/a
Kick→Loco	random tick, incl. strike	98.8	n/a	n/a	n/a
Loco→Kick	from $v \approx 0$ state	99.3	95.3	17.0	32.7
Loco→Kick	in motion, 25% range	97.2	90.2	14.7	29.8
Loco→Kick	in motion, 50% range	97.0	87.8	13.7	27.5
Loco→Kick	in motion, full range	95.8	84.1	11.9	25.7

TABLE IX: Hardware results.

(a) Shooting trials.					
Condition	Trials	Suc. [†] (%)	Err. (m)	Hit (%)	Alive (%)
Static ball, forward	32	46.9	0.25	93.8	100.0
Static ball, lateral	30	53.3	0.19	90.0	100.0
Static ball, rearward	30	20.0	0.87	93.3	96.7
Commanded approach	30	40.0	0.30	93.3	100.0

[†]Radius 0.14 m, i.e. $\pm 5.7^\circ$ at 1.42 m, not the simulated 0.5 m.

(b) Handoff robustness, Kick→Loco, manual abort (20 trials each).

Abort phase	Pre-strike	Mid (contact)	Post/recovery
Alive (%)	95.0	95.0	100.0

TABLE X: Ablations on skill_7, medians over eight matched Stage-2 checkpoints per arm.

(a) Post-strike stabilisation mechanism.				
Arm	Stance asym.↓ rad ²	Yaw drift↓ (rad/s) ²	Orient. tilt↓ sin θ	
Posture + contact safety (ours)	0.013	0.012	0.003	
Contact safety only	0.276	0.074	0.163	
No stabilisation	0.117	0.044	0.073	
(b) Kick-state pose injection for locomotion recovery.				
Training arm	Alive, ref.-clip (%)	Alive, live donor (%)		
Transplant + ref.-clip (ours)	88.4	81.8		
Reference-clip pose only	82.3	59.7		
No pose injection	71.8	32.1		
(c) Locomotion-mode tracking gate.				
	v_x (m/s)		v_y	ω_z
Tracking MAE↓	fwd.	back.	(m/s)	(rad/s)
Hard zero (ours)	0.34	0.46	0.35	0.14
Floor 0.1	0.38	0.58	0.37	0.15
Floor 0.5	0.67	0.79	0.40	0.17

D. Locomotion generality

Table XI is the axis prior systems do not report: paired against the same substrate trained without any skill layer, the claim is an absence of effect: attaching seven skills should not degrade locomotion. We retrained the substrate under two further seeds to give that claim a noise floor: every Δ sits inside one standard deviation of the baseline’s own three-seed spread, and all but backward tracking inside half of it. Leaving the library side at one seed only understates that floor, but it cannot exclude a small degradation reproducing across seeds: all four tracking deltas are non-negative. Terrain traversal uses the same sustained-fall criterion training itself uses (10 consecutive ticks below threshold, not an instantaneous dip).

VII. LIMITATIONS AND CONCLUSION

A. Limitations

Distillation loses most of the teachers’ accuracy, and the loss grows with library size. This is the system’s

TABLE XI: Locomotion generality with the skill library attached. Both policies build on the seed-42 substrate, and Δ is that paired difference, computed before rounding. The s.d. row is the spread of three Stage-1 seeds (42/7/13), a noise floor rather than an interval around the row above: their terrain rates are 98.7/97.3/62.7.

	v_x MAE		v_y	ω_z	Terr.	Push
	fwd.	back.	(m/s)	(rad/s)	(%)	(%)
Loco-only, seed 42	0.39	0.51	0.35	0.21	98.7	100.0
s.d. over 3 seeds	0.07	0.05	0.07	0.17	20.4	0.0
With library	0.42	0.55	0.35	0.25	100.0	100.0
Δ	+0.02	+0.04	+0.00	+0.04	+1.3	+0.0

principal open problem (Table VII). Mean success at 0.5 m falls from 33.2% across teachers to 15.4% merged. Whether this is a capacity limit or a fixable distillation artifact is open: skills move non-monotonically across library sizes, and the distillation has no seed replicates. Our multi-skill claim is thus about coverage and stability, not specialist accuracy. The gate ablation is likewise one skill at Stage 2, one seed. It shows that discounting tracking costs command tracking, not that the hard zero is required at library scale.

The strikes are both mis-aimed and imprecise. Across 112 checkpoint-skill regressions of the ball’s post-contact bearing on the commanded offset θ , per-skill median slopes span 0.02 to 0.88 rather than 1, and one fit in five is negative, steering the wrong way. Each fit also carries an angular offset, up to 40° by skill and unstable across checkpoints. Per-trial scatter about that offset is $9\text{--}24^\circ$, wider than the $\pm 5.7^\circ$ a 0.5 m hit allows. Both terms therefore bound the shot-error and success columns of Tables V and VI. Correcting the offset recovers much of this (Table XII): refitting each skill’s nominal bearing to its measured departure bearing, cross-validated within each checkpoint, improves success at both radii by $1.7\text{--}1.8\times$ over the same trials scored without it, on all 16 checkpoints (sign test $p < 10^{-4}$) and every skill. The command is a relative offset, so this is what a recalibrated deployment would measure, not a projection. The offsets replicate across halves of a checkpoint but not across checkpoints, so the fit is per deployed checkpoint, not once per skill.

TABLE XII: Per-skill angular calibration, medians over Table VI’s 16 checkpoints, on a separate 250-trial scan. δ is fitted on half of each checkpoint’s trials and scored on the other, then swapped.

Skill	1	2	3	4	5	6	7	All
δ (deg)	+21	−40	−8	−11	−5	−17	0	−
Suc.@0.5 (%)	10.4	1.2	21.8	26.8	36.2	8.6	9.6	16.4
+ calibration	28.8	20.2	29.0	41.6	40.4	27.4	16.8	29.2
Suc.@1.0 (%)	22.4	4.8	38.8	52.2	65.2	24.0	20.2	32.5
+ calibration	56.0	42.4	49.0	71.6	74.8	51.2	33.6	54.1

B. Conclusion

We presented a humanoid soccer policy built the other way around from current practice. A commandable locomotion policy is trained first and alone, with N motion-guided kicking skills layered on as task-gated additions held at

exactly zero outside kick mode. The hard-zero gate fixes the reachable gait space to the command curriculum, not to any reference clip. And because every skill begins and ends in the same locomotion state, a library of N skills needs $O(N)$ transitions, not $O(N^2)$.

On the metrics prior systems report, the locomotion-grounded specialist reaches 44.3% success at 0.5 m, over a narrower aim cone and at lower pace than prior work. The distilled library averages 15.4% across the seven bearings, at a locomotion cost of at most 0.04 in command-tracking MAE, and transfers to hardware.

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